

Effects of Delivery Time and Delivery Distance of Indoor Robot Delivery Service on User Satisfaction and Reuse Intention

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ABSTRACT

The indoor robot service industry has been growing rapidly, industrial and academic research has been actively conducted. However, several previous studies have focused on the acceptance of robots, and there is a lack of research on delivery robot services. Moreover, research cases targeting users who have experienced actual robot delivery services are rare. Therefore, we conducted this study targeting employees who have used actual robot delivery services in a large office space with a total floor area of 165,000 m² built as a robot-friendly building. Providing delivery services with approximately 100 robots in a large building is a rare case in the world. This study expanded the technology acceptance model and analyzed how delivery time and distance, which are the characteristics of robot delivery services, affect the robot acceptance intention.

CCS CONCEPTS

• Human-centered computing; • Human computer interaction (HCI); • Empirical studies in HCI;

KEYWORDS

Service robots, Robot delivery service, Technology adoption

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1 INTRODUCTION

With the recent application of new technologies (artificial intelligence, 5G, etc.) of the Fourth Industrial Revolution to robots, robots have become increasingly intelligent, and various types of robots, such as 5G-based cloud robots and indoor/outdoor delivery robots, have emerged. According to the International Federation of Robotics, in the 2020 worldwide robot market, the service robot industry grew by 13% year-on-year owing to the increases in the medical robot (+174%), professional cleaning robot (+95%), and indoor delivery and logistics robot (+33%) sectors. The increase in the aging population, the proliferation of non-face-to-face interactions in

society owing to the COVID-19 pandemic, and the advancement of AI technology is further promoting the development and spread of service robots [6, 7, 15, 18]. With the growth of the robot market, studies on service robots and autonomous delivery robots are actively underway, and several researchers are investigating the factors that affect the robot use intention and acceptance intention when users encounter robots [1] - [9].

Previous studies on the acceptance of service robots focused on various aspects, such as social impact, emotion, anthropomorphism, and pleasure [1, 8, 9]. However, there is a lack of research on how delivery time and distance, which are the characteristics of the delivery of robot services to users, affect robot acceptance. Therefore, to overcome the aforementioned limitations of previous studies, we conducted this study based on the characteristics of delivery robot services and the value provided to users by expanding the technology acceptance model (TAM). This study targeted people who had used delivery robot services.

We reviewed previous studies on robot use intention and found that several studies were conducted on users who had not experienced service robots, potential future users, or users with indirect experiences [2, 3, 5]. As delivery service robots have not yet become commonplace, surveys were conducted with users who had yet to experience robots in previous studies, which is a limitation of previous studies. Recently, Naver, a giant information technology (IT) company in South Korea, has been providing robot delivery services with approximately 100 robots for a 28-story building with a total floor area of 165,000 m². Therefore, this study aims to investigate the impact of the actual delivery robot service on the satisfaction and reuse intention of employees who use the delivery robot service.

2 RELATED WORK

2.1 Robot acceptance model

The TAM has been typically used in research as a behavior model of users accepting a new technology when it emerges. Davis showed that perceived usefulness (PU) and perceived ease-of-use (PEOU) in information systems are fundamental factors in user acceptance [10, 11]. The TAM has been well established as a powerful and simple model for predicting user acceptance [12]. Empirical research based on the technology acceptance theory has been conducted in various fields. However, the TAM has limitations in understanding the impact on the technology acceptance of users from various perspectives. Therefore, the model was expanded to TAM2 by adding external factors affecting PU, and to TAM3 by reflecting six factors affecting PEOU (computer self-efficacy, perception of external control, computer anxiety, computer playfulness, perceived enjoyment, and objective usability).

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Studies have been actively conducted to mitigate the limitations of the TAM and investigate the factors affecting the robot use intention and acceptance. Based on TAM3, Bröhl et al. analyzed the robot acceptance intention when robots and humans work in collaboration in a production system. Stock and Markle combined the role and TAM theories to develop a social frontline robot acceptance model [9]. Heerink et al. expanded research on the robot acceptance based on TAM and the unified theory of the acceptance and use of technology and proposed a new model for assistive social robots by adding the perceived enjoyment, perceived sociability, and social presence, and perceived adaptability factors [16]. Go et al. emphasized that perceived interactivity in machine-learning-applied robots is greater in the robot acceptance model and proposed an interactive technology acceptance model [1].

2.2 Delivery time and distance

When users perceive that a new system can save more time and effort, they have a positive response and use intention [23]. Several previous studies on services and marketing have focused on the effect of waiting time on satisfaction with the service. Waiting is one of the most important factors in the service experience [25]. Waiting time can sometimes evoke positive emotions, such as anticipation or hope, but most customers experience various degrees of discomfort, such as irritation, boredom, and frustration, while waiting [25]. Therefore, waiting is associated with a decrease in the perceived quality of the service, service failure, and customer dissatisfaction [27]. Studies analyzing the difference between perceived and actual waiting times argue that perceived waiting time has a greater impact on consumers' waiting perception, experience rating, and behavior than actual waiting time [24]. Larson et al. (1991) argued that, as it is impossible to eliminate waiting time completely, efforts should be focused on methods of managing the customer perception of waiting time [21]. Robot delivery services are differentiated in that humans are replaced with robots for the delivering role in the same process as existing online food delivery (OFD) services. Several studies in the OFD field argue that one of the most important motivations of people for using technology-based self-services is the intention to save time [37]. In studies conducted from the perspective of comparing human and robot acceptance, people tend to evaluate service robots based on the service provided by humans as a reference point [9]. Although humans and robots are vastly different, customers want to receive services at a level similar to that of human services. Therefore, if customers feel that service robots are inferior in performance compared with humans, they may not prefer robot services [39]. In the field of logistics, too, several studies have been conducted on delivery time. Customers increasingly want improvements in the quality of logistics services, particularly the shortening of delivery time. Several customers think that they are "short of time," and a long delivery time makes customers feel anxious and dissatisfied [28]. In a study on delivery distance, it was observed that customers who need to walk more than 10 min, or more than 0.5 miles from a store, are more willing to pay for autonomous delivery robots [18]. Based on the aforementioned previous studies, this study empirically analyzes the effects of the perceived delivery time, actual delivery time, and

actual distance of the robot delivery service on the robot service acceptance.

3 METHODS

In this study, we examine how the travel time and distance of delivery robots that have replaced humans affect the intention to use robots. Therefore, we conducted a study targeting employees who had operated and used actual services in a large office building. The robot delivery service provided in the indoor office performed physical travel on behalf of employees and saved the time spent on travel. We aimed to demonstrate empirically the effects of not only the function of the robots but also the delivery time factor, which affect the entire experience from the beginning to the end of the service, on the robot acceptance. Furthermore, we examined the difference in using the delivery service between the users on physically close low floors where the access to stores was relatively easy and the users on high floors. The research questions are as follows:

RQ1. How will the actual delivery time and perceived delivery time of the robot delivery service affect the intention to use the robot service?

RQ2. How will the delivery distance between the store and the customer in a large office affect the intention to use the robot service?

The PU and PEOU of the traditional TAM are useful tools for explaining the TAM. In this study, the delivery time and distance, which will represent the characteristics of the delivery robot service together with the TAM, were added as independent variables.

H1. The shorter the perceived robot delivery time, the greater is the positive impact on user satisfaction and reuse intention.

H2. The shorter the actual delivery time, the greater is the positive impact on user satisfaction and reuse intention.

H3. The longer the actual delivery distance, the greater is the positive impact on user satisfaction and reuse intention.

3.1 Environments and service scenario

Naver, South Korea's leading IT company, built a new company building to be robot-friendly and built a large-scale robot service testbed (Figure 1). All the facilities in this building, such as automatic doors, gates, and elevators, are linked to the multirobot control platform and facilitate horizontal and vertical travels of robots when moving. Approximately 5,000 employees work in this building, and approximately 100 robots provide services of delivering coffee, food, and parcels to the desks of employees [32–34]. The employees can use the robot delivery service at any time they want from 10:00 a.m. to 6:00 p.m. on weekdays.

Figure 2 shows a scenario for the café robot delivery service. Step 1. An employee orders a beverage through the mobile app at his/her desk in the company. Step 2. The order is forwarded to the POS device, and the barista who sees the order makes the beverage, calls a robot, and then loads the beverage onto the robot. Step 3. The robot travels by elevator to the destination. Step 4. The employee sees the robot arriving near his/her desk and picks up the beverage.



Figure 1: NAVER robot-friendly building. source of the picture (<https://1784.navercorp.com>)

3.2 Participants and data collection

In this study, we sent a questionnaire to all 636 employees who used the café robot delivery service at least once from September to October, 2022. Among the 636 employees who received the questionnaire by email, 220 responded to the survey (35% response rate). The gender composition was 136 males (62%) and 84 females (38%). The participating age groups consisted of 58 people in their 20s (26%), 108 people in their 30s (49%), 50 people in their 40s (23%), and 4 people in their 50s (2%). The data in this study included the satisfaction, reuse intention, PU, PEOU, and perceived delivery time collected from the survey responses. Moreover, the actual robot delivery time and the ordered building floor information extracted from the service log files were added (see Table 1).

4 RESULTS

In this study, we performed multiple regression analyses to test the hypotheses (See Table 2). In the regression model F-test of each model, the probability of significance is $p < 0.01$, which indicates that the regression model is suitable. The independence of the residuals was determined by the Durbin–Watson statistic, which was

2.276 for Model 1 and 2.244 for Model 2, indicating that the independence of the residuals was satisfied. The adjusted R^2 value had explanatory powers of 0.385 (Model 1) and 0.433 (Model). From the analysis results, it cannot be said that gender and age have significant effects on satisfaction and reuse intention. First, upon examining the analysis result for the delivery time, it is observed that, in Model 1, the perceived delivery time ($B = 0.238$, $p < 0.01$) has a positive effect on user satisfaction. However, the actual delivery time has no significant impact on user satisfaction and reuse intention. In terms of delivery distance, as the actual delivery distance ($B = 0.091$, $p < 0.1$) increases, i.e., in the case of employees working on high floors, there is a positive impact on the intention to reuse the robot delivery service. The delivery store is located on the second floor, and employees working on relatively high floors, thus having long physical travel distances, have higher reuse intention than those working on low floors. The PU ($B = 0.340$, $p < 0.01$) and PEOU ($B = 0.239$, $p < 0.01$) of the traditional TAM also show that there are positive impacts on user satisfaction and reuse intention for the robot delivery service, similar to previous studies.

After examining the regression analysis results, in which the actual delivery time had no significant impact, we examined the minimum, maximum, and mean of the actual delivery time for each perceived delivery time (see Table 3). The actual delivery time for people who felt that the perceived delivery time was “a little slow” was 6 min and 39 s, and the minimum actual delivery time for those who rated the service “very fast” was 6 min and 59 s. In other words, there were cases where the actual delivery time was shorter for those who rated the service “a little slow” than those who rated it “very fast.”

5 DISCUSSION

In this study, we empirically analyzed the effect of the delivery distance and time on user satisfaction and reuse intention for employees using a café robot delivery service in a large office building.

First, in the aspect of delivery time, we analyzed the effects of the actual delivery time and perceived delivery time on user satisfaction and reuse intention. The perceived delivery time had a

Table 1: Measured items

Variable	Measure	Definition
Satisfaction	5-point Likert scale	The degree to which the respondent is satisfied with using the robot delivery service [14]
Intention to Reuse	5-point Likert scale	The degree to which the respondent intends to reuse the robot delivery service [16]
Perceived Usefulness	5-point Likert scale	The degree to which the respondent believes that the robot delivery service will improve or help him/her perform his/her work [9, 10]
Perceived Ease of Use	5-point Likert scale	The degree to which the respondent feels that the robot delivery service is easy and comfortable to use [9, 10]
Perceived Delivery Time	5-point Likert scale	The degree to which the respondent feels that the delivery is fast/slow from ordering to receiving through the robot delivery service [30]
Actual Delivery Time	Time (s)	The actual time it took to receive after ordering through the robot delivery service [31]
Actual Delivery Distant	Floor	The actual distance traveled for the robot delivery service

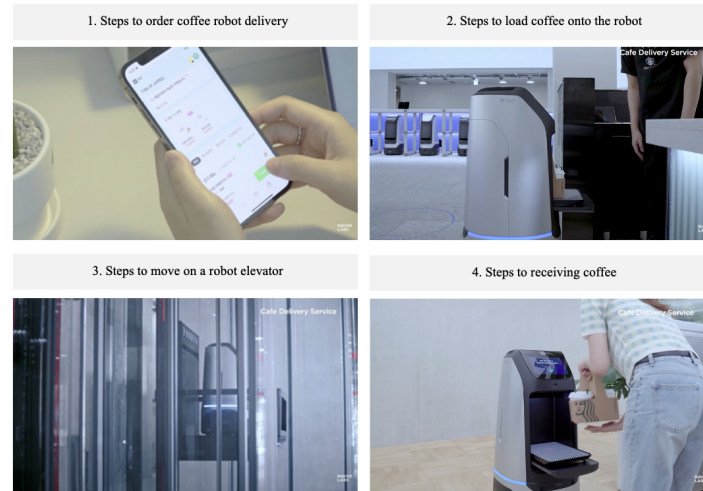


Figure 2: Naver robot delivery service process: source of pictures (<https://youtu.be/6eGtl5LQpf8>)

Table 2: Multiple regression analysis

	Model 1 Dependent Variable: User Satisfaction	Model 2 Dependent Variable: Intention to Reuse
	Standardized Beta	Standardized Beta
Sex (male: 0, female: 1)	-0.098	0.011
Age (10s: 1, 20s: 2, 30s: 3, 40s: 4, 50s: 5)	0.004*	-0.022
Perceived Usefulness	0.340***	0.522***
Perceived Ease of Use	0.239***	0.226***
Perceived Delivery Time	0.238***	0.043
Actual Delivery Time	0.048	0.004
Actual Delivery Distant	0.034	0.091*
F	20.607***	24.880***
R^2	0.405	0.451
Adjusted R^2	0.385	0.433

* $p < 0.1$. ** $p < 0.05$. *** $p < 0.01$

Table 3: Perceived delivery time vs. Actual delivery time

Perceived Delivery Time		Actual Delivery Time		
Measures	N	Minimum	Maximum	Mean
1 - Very Slow	15	0:07:14	0:22:58	0:11:26
2 - Slightly Slow	32	0:06:39	0:22:16	0:09:57
3 - Normal	80	0:06:44	0:22:57	0:09:16
4 - Slightly Fast	60	0:06:38	0:15:55	0:08:56
5 - Very Fast	33	0:06:54	0:12:06	0:08:33

significant effect on user satisfaction. In other words, user satisfaction improved when users felt that the delivery time was fast in their subjective judgment. However, the actual delivery time had no significant effect on user satisfaction and reuse intention. Even though the actual delivery time was longer, there were users who felt that the perceived delivery time was shorter. Conversely, some users felt that the perceived delivery time was longer even though

the actual delivery time was shorter. Comparing the effects of perceived delivery time and actual delivery time, we observed that the perceived delivery time has a greater effect on user satisfaction than the actual delivery time. This is similar to the result of a previous fast-food restaurant case study, in which the perceived waiting time had more significant impact on waiting perception than the actual waiting time. This result indicates that, when designing a

robot delivery service, the functional factors that can reduce the perceived delivery time need to be examined to improve user satisfaction. If factors that can reduce the perceived waiting time when waiting for a robot delivery service are improved, they will have positive effects on satisfaction with the robot delivery service and the reuse intention. Some previous studies of the restaurant service domain have observed that the service environment (lighting, color, temperature, music, interior design, etc.) can reduce the perceived waiting time [35, 36]. With the robot delivery service, one can perform desired work or leisure activities without moving from his/her location. Therefore, it is expected that the perceived waiting time will be reduced compared to waiting in line at a restaurant. Second, we empirically analyzed whether there is a difference in the effect on reuse intention between users working on low floors and those working on high floors where the physical travel distance and time can be reduced by the robot delivery service. The level of the floor had no significant effect on satisfaction with the robot delivery service, but it had a positive effect on the reuse intention. This is the first study to empirically analyze the satisfaction with the robot delivery service and the reuse intention for customers working on high floors in a large office building. This indicates that, when selecting a target building for an indoor robot delivery service from a practical point of view, it is necessary to consider a high-rise building or a place where the distance between the store and the workspace is long. However, as the number of samples for each floor needs to be increased, and teams or organizations working on each floor are different, further research should be conducted to generalize the results. Finally, the PU and PEOU, which are independent variables of the traditional TAM, were at a significance level of 0.01 for satisfaction and reuse, confirming positive effects. As demonstrated empirically in several previous studies, the PU and PEOU have significant effects when users accept technology in the case of robot delivery services.

6 CONCLUSION

The perceived delivery time has a positive effect on user satisfaction. As the actual delivery distance increases, it has a greater positive effect on the intention to reuse the robot delivery service. However, the actual delivery time has no significant impact on user satisfaction and reuse intention. The contributions of this study are as follows. First, we expanded a model reflecting the service characteristics of delivery robots from the TAM and validated it. Several previous studies focused on the acceptance model of robots. However, this study analyzed technology acceptance factors from the perspective of distance and time, which are important factors when robots and services are combined, particularly in a delivery service. This indicates that when designing a delivery robot service, not only the robots themselves but also the traveling routes of robots, the delivery time perceived by users, and the service environment should be considered important. Second, we performed an analysis based on actual users who had used robots for personal needs instead of using them for experiments. Previous studies targeted potential users who had not used robots when verifying robot acceptance models, and they were conducted through users responding to a questionnaire after looking at pictures of robots or thinking about using them in the future. However, this study derived results from

the actual measured data of a robot delivery service. Third, this study performed an empirical analysis with a robot delivery service in a large office building. The new company building of Naver, the target of this study, is a huge robot test building where robot services are operated constantly on a large scale with approximately 100 robots. There are very few cases where actual robot services are operated in a large office space, and there also is a lack of previous studies on such a case. Our study will provide practical implications to companies providing robot services in similar large buildings.

A limitation of this study is that service factors and additional factors of existing robot acceptance models were not included more comprehensively. If we add more comprehensive influential factors to perform analysis in future research, we will be able to derive more significant research results. Another limitation is that this study targeted the employees of only one company, and the target statistical sample is not representative of all companies. Nevertheless, considering that the robot service market is still in the infancy stage and there is a lack of environments where robot delivery services of this magnitude can be experienced, it is determined that this study is significant. In the future, if robot delivery services spread in the market, and research will be frequently conducted for broader targets, better insights can be derived.

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